

Edge Board System Installation Guide

Purpose

This guide replaces the legacy installation note with an English operator checklist for preparing the edge compute board used by the dual-arm box-opening system.

Required Equipment

- Host computer running a supported Linux desktop release.
- Edge compute board, power adapter, storage media, and data cable.
- Network connection for package installation and repository access.
- Display, keyboard, and mouse for first-boot validation when needed.

Installation Steps

- Download the correct board image and flashing tools for the hardware revision.
- Put the board into recovery or flashing mode according to the hardware manual.
- Copy the flashing command into a plain-text note, check all spaces and line breaks, then execute it from the host computer.
- Wait for flashing to complete and allow the board to reboot automatically.
- Complete the first-boot account, network, timezone, and storage configuration.
- Install terminal utilities, development tools, camera utilities, and Python package tooling for the board architecture.

Post-Install Validation

- Confirm network connectivity and package repository access.
- Confirm available disk space and memory.
- Confirm RGB-D cameras are visible to the operating system.
- Confirm the robot controller network is reachable from the board.
- Record the final system image version and keep it with the deployment log.

Next Step

After the base system is ready, use the Ubuntu 22.04 ROS 2 Humble and MoveIt2 deployment document for the recommended robotics environment.